

SiagaKita: Integrated Tourism Emergency Response & Real-Time Tracking Ecosystem

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Abstract. The increasing complexity of emergency situations and the vulnerability of tourism areas to disaster risks require more integrated, responsive, and technology-driven emergency response systems. This study aims to design and develop SiagaKita, a mobile-based platform that integrates communities, volunteers, and government agencies into a real-time multi-actor coordination ecosystem. The research employs a Research and Development (R&D) approach with qualitative methods, including in-depth interviews, observations using the AEIOU framework, and literature review. Data were analyzed using thematic analysis to identify patterns of needs, challenges, and interactions within the emergency response system. The findings reveal that key issues include fragmented communication, delayed response due to lack of data integration, and users' limitations in providing accurate information during emergencies. The proposed system integrates GPS-based automatic reporting, incident classification, real-time tracking, and multi-actor coordination, which significantly enhance information accuracy and response efficiency. Furthermore, the integration of user data and the involvement of volunteers strengthen early response capacity. This study concludes that an integrated mobile-based system offers significant potential to improve emergency response effectiveness and provides important implications for policy development and digital system innovation in tourism and disaster management sectors.

Keywords: emergency response, integrated system, mobile application, real-time coordination, disaster management

1. Introduction

The escalating intensity and complexity of natural disasters and global emergencies over the past several decades have positioned emergency response systems as a crucial component in safeguarding human safety and sustaining socio-economic activities, including the tourism sector, which is particularly vulnerable to various environmental and non-environmental risks [1]. Digital transformation through mobile technology, the Internet of Things (IoT), and real-time data-driven systems has opened new opportunities for enhancing disaster management effectiveness, especially in communication, coordination, and critical information distribution [2]. In the national context, Indonesia as a country with high disaster vulnerability faces serious challenges in managing tourist safety, particularly at nature-based destinations with significant geological and hydrometeorological risks. This situation demands innovation in emergency response systems that are not only responsive but also adaptive to regional characteristics and user behavior.

Although various emergency response systems have been developed, both in mobile applications and digital communication platforms, research indicates that significant gaps remain in system integration, data accuracy, and stakeholder coordination [3]. Conventional systems relying on manual reporting and fragmented communication often lead to delayed responses, miscommunication, and inefficiency in mobilizing emergency resources [4]. Furthermore, challenges such as the spread of unverified information, lack of report validation, and limited network access at tourist sites further undermine the effectiveness of existing systems [5]. These gaps highlight the academic

and practical urgency of developing an emergency response system model capable of integrating multiple actors, leveraging real-time data, and simultaneously improving accuracy and response speed.

Conceptually, this study is grounded in the technological mediation approach within disaster risk management, which positions technology as the primary infrastructure facilitating information exchange, multi-actor coordination, and data-driven decision-making [6]. The concept of real-time emergency communication systems emphasizes the importance of situational visibility and information integration in improving emergency response efficiency [7]. Another relevant approach is ICT-enabled co-production, where community and volunteer involvement in technology-based emergency response systems can accelerate initial response and strengthen community resilience [8]. The integration of these concepts forms the foundation for designing a system that functions not merely as a reporting tool, but as a comprehensive coordination ecosystem.

Based on these identified gaps, this study aims to design and develop an integrated mobile-based emergency response system, SiagaKita, capable of connecting communities, tourists, volunteers, and official agencies on a single real-time coordination platform. Specifically, this study seeks to address: (1) how users' and agencies' needs are characterized within emergency response systems in tourism areas; (2) how to design a system architecture capable of improving reporting accuracy and coordination efficiency; and (3) how the effectiveness of a multi-actor system enhances emergency response. The approach employed is a qualitative R&D method integrating primary and secondary data to produce an applicable and contextual system model.

The scientific contribution of this study lies in the development of an integrated multi-agency emergency response system model that combines mobile technology, real-time tracking, and user data integration within a unified ecosystem. Unlike previous research tending to focus on a single aspect such as emergency reporting or early warning systems, this study offers a holistic approach integrating reporting, verification, incident classification, and cross-institutional coordination simultaneously [9]. Additionally, this research enriches information systems scholarship by combining technological, user behavior, and operational perspectives within one comprehensive design framework, serving as a reference for digital emergency response system development in tourism and disaster-prone regions.

The conceptual foundation of this study is rooted in technology-based disaster risk management theory, which emphasizes the role of information and communication technology (ICT) as the primary mediator in coordination, information distribution, and decision-making during emergencies. From this perspective, technology functions not merely as a supporting tool but as a core infrastructure shaping actor interactions within emergency response systems [6]. Additionally, real-time emergency response systems concepts stress the importance of speed, accuracy, and data connectivity in mitigating disaster impacts, where the integration of GPS, cloud computing, and mobile systems enables significant improvements in situational visibility [1]. IoT and real-time data analytics further enable early detection, environmental monitoring, and automatic alerts to relevant users and agencies [10]. The integration of technology, data, and actors thus forms the primary foundation for developing modern emergency response systems.

Several prior studies have developed mobile-based systems to improve emergency response effectiveness, both for natural disasters and accidents. Research on Android-based safety applications demonstrates that integrating SOS features, location tracking, and real-time communication can significantly improve response speed compared to conventional methods [9]. Other studies have developed machine learning-based systems capable of predicting disaster potential and issuing early warnings to users, thereby enhancing community preparedness [11]. Additionally, research on relief coordination platforms shows that integrating users, volunteers, and agencies through a centralized system reduces miscommunication and accelerates aid distribution [3]. Other studies highlight the importance of crowdsourcing and artificial intelligence in improving data accuracy and decision-making efficiency in emergency situations [5].

Nevertheless, a literature review reveals several research gaps that remain inadequately addressed. One key weakness in existing systems is the lack of cross-institutional integration, resulting in coordination between agencies such as fire departments, medical services, and disaster management bodies operating in silos [4]. Furthermore, most systems focus on reporting or early warnings without providing real-time feedback mechanisms to users regarding the status of ongoing assistance [12]. Another limitation identified is the lack of integration of personal user data such as medical information which is critically important in determining initial treatment at the scene [13]. Additionally, data verification challenges and fake information filtering remain crucial issues in crowdsourcing-based systems [14].

In this context, this study positions itself as an effort to address the integration gap in emergency response systems by developing a multi-agency real-time emergency response model that integrates not only reporting but also cross-actor coordination and comprehensive user data management. This approach extends prior studies by combining several key components, such as automatic incident classification, real-time tracking, and biometric user data integration within a unified platform. The study also adopts a user-centered design approach emphasizing ease of use in emergency conditions, as recommended in mobile application-based emergency response research [15]. Thus, this study contributes both technically and conceptually to expanding the paradigm of technology-based emergency response systems.

Recent research trends indicate a shift toward more integrated and data-driven approaches in emergency response system development. Technologies such as AI, IoT, and cloud computing are increasingly dominant in enhancing prediction, analysis, and coordination capabilities [16]. Geospatial systems and 3D visualization are also being developed to improve situational awareness and emergency response planning [17]. Methodologically, many studies adopt agile system design and user-based evaluation approaches, enabling more adaptive solutions responsive to field conditions [12]. The integration of volunteers as part of ICT-based emergency response systems also represents an important trend in building response capacity and community resilience [8].

Based on the overall literature review, a conceptual synthesis can be drawn: effective emergency response systems must meet several key principles, including multi-actor integration, real-time data availability, information accuracy, and adaptive capacity to user and environmental conditions. This approach demands a system architecture capable of connecting diverse data sources and actors within a coordinated platform while providing continuous verification and feedback mechanisms. By combining technological mediation, real-time communication, and ICT-enabled coordination concepts, this study builds a conceptual framework underlying the design of SiagaKita as an innovative solution for improving digital-based emergency response management effectiveness.

2. Research Method

This study employs a Research and Development (R&D) approach with a qualitative paradigm, aimed at designing, developing, and validating an integrated mobile-based emergency response system. The R&D approach was selected because it enables researchers to not only understand empirical phenomena in the field, but also generate applicable solutions based on the needs of users and relevant agencies. Epistemologically, the qualitative approach is used to explore the experiences, perceptions, and needs of actors in the emergency response system in depth, yielding contextual understanding not achievable through a purely quantitative approach [18]. This study's strategy also adopts an exploratory-descriptive approach focused on problem identification, needs analysis, and information systems-based solution design.

Data sources consist of primary and secondary data. Primary data were obtained through in-depth interviews and direct observations of various actors in the emergency response ecosystem, including general community members (users), volunteers, and official agencies such as BPBD, BASARNAS, and the fire department. Secondary data were obtained through a literature review covering scientific journals, policy reports, and documents related to disaster management systems and emergency response technology. The combination of primary and secondary data enables information triangulation to enhance the validity of research findings, as recommended in contemporary qualitative approaches [19].

Data collection employed three main methods: in-depth interviews, field observations, and document studies. Interviews were conducted semi-structurally to allow flexibility in gathering in-depth information about respondents' experiences and needs in emergency situations. This technique is considered effective in qualitative research as it produces rich and contextual data [20]. Observations were conducted using the AEIOU (Activities, Environment, Interaction, Objects, Users) framework to understand user behavior and interaction dynamics in real contexts, generating more comprehensive insights into system requirements. Document studies supplemented the data with secondary information relevant to the research topic.



Figure 1. Interview with BASARNAS representatives during the data collection process

Inclusion criteria encompassed participants with direct experience or involvement in emergency situations whether as victims, volunteers, or agency personnel—and scientific literature relevant to technology-based emergency response systems published within the past five years. Exclusion criteria included data not relevant to the emergency response context, sources lacking academic credibility, and literature not providing open access. These criteria aim to maintain the quality and relevance of data used in the analysis process and ensure that research results are based on valid and scientifically accountable information.

The unit of analysis in this study comprises actors within the emergency response ecosystem, classified into four main groups: end-users (general community and tourists), first responders (volunteers), emergency response agencies (official institutions), and system administrators. This selection is based on each actor's strategic role in the emergency response chain, enabling comprehensive analysis of needs, barriers, and inter-actor interactions. This approach aligns with qualitative research principles emphasizing understanding the experiences and perspectives of research subjects within their social context [21].

Data analysis employed Thematic Analysis to identify user needs, emergency response handling problems, and solutions translated into SiagaKita application features. Analysis proceeded through stages of data familiarization, manual coding, theme grouping, and interpretation of interview, observation, and documentation results. The derived themes served as the basis for compiling system requirements, feature design, user flows, and application interface design before implementation using the Flutter framework. This approach was selected for its capacity to produce systematic analysis oriented toward user needs, ensuring that developed features align with actual emergency handling conditions [22]. The study also considers data saturation as an indicator of data sufficiency when no new themes emerge from the collection process [23].

3. Results

This section presents the main findings obtained through qualitative data collection and analysis, based on in-depth interviews and field observations with actors in the emergency response ecosystem. Overall, findings indicate that current emergency response systems still face various structural limitations, particularly regarding data integration, response speed, and inter-agency coordination. Through thematic analysis, the obtained data were categorized into several major themes representing patterns of needs, barriers, and opportunities for technology-based system development. Thematic analysis enables the identification of recurring patterns in data, subsequently classified into systematic thematic categories [24].

The first finding relates to fragmentation in reporting and emergency coordination systems. Based on interviews with representatives from agencies such as BPBD and BASARNAS, community reports were found to frequently lack standardization, precise location coordinates, and were dispersed across multiple non-integrated communication channels. This caused delays in the verification process and aid mobilization. Field observations also showed that coordination still relies on manual communication—such as telephone calls or instant messaging—

prone to miscommunication. This pattern indicates that the existing system has yet to establish an integrated common operating picture, as recommended in modern emergency management system development [25].

The second finding identifies an information gap regarding victims' physiological conditions. Interview data from field officers revealed that in many cases, medical intervention was performed without prior information about the victim's health status, such as medical history, allergies, or other special needs. This increases the risk of mishandling, particularly in critical situations requiring rapid decisions. Furthermore, the absence of a system capable of integrating personal user data with emergency response systems renders the handling process suboptimal. This finding highlights the importance of user data integration in technology-based systems to improve medical intervention accuracy [13].

The third finding relates to users' limited capacity to convey information during emergencies. Based on AEIOU framework observations, it was found that in panic conditions, users tend to experience diminished cognitive capacity to describe location, incident chronology, and assistance needs in detail. This was reinforced by interview results with tourists and small business operators who reported difficulty accessing evacuation information and emergency services. This finding indicates that emergency response systems need to be designed with user-centered design principles enabling simple, fast, and minimally complex reporting, as recommended in mobile application-based safety system development [9].

The fourth finding reveals a need for real-time multi-actor coordination systems. Data indicates that volunteer involvement as first responders has significant potential in accelerating initial response, particularly within a specific geographical radius before official assistance arrives. However, the absence of a system capable of systematically coordinating volunteers means this potential has yet to be optimally utilized. Field studies also showed that volunteers often operate independently without coordination with official agencies, potentially causing duplication or task conflicts. This finding indicates the importance of integrating volunteers into ICT-based systems to improve emergency response efficiency [8].

The fifth finding relates to victims' need for information transparency and visibility. Interview results indicate that uncertainty regarding assistance status is one factor increasing victim anxiety in emergency situations. Users desired a system capable of providing real-time information about assistance position, estimated time of arrival (ETA), and handling status. This shows that beyond operational aspects, emergency response systems also have a psychological dimension requiring attention. Real-time tracking-based systems have been shown to enhance user trust and reduce stress levels in crisis situations [1].

Overall, these findings confirm that the primary needs in emergency response systems encompass data integration, information accuracy, ease of use, and real-time multi-actor coordination. These findings address the research questions previously formulated—regarding user and agency needs, effective system design, and the role of technology integration in improving emergency response. No rejection of the study's foundational assumptions was found; rather, findings reinforce the importance of developing an integrated technology-based emergency response system capable of addressing the complexity of field needs [26].

4. Discussion

The findings of this study indicate that the primary problems in emergency response systems lie in information fragmentation, delayed responses, and lack of inter-actor integration, which can be conceptually explained through the technological mediation framework in disaster risk management. From this perspective, technology serves as the primary connector in building cross-actor coordination and creating situational visibility in real time. Results showing the need for integrated real-time data systems align with the principle that emergency response effectiveness depends heavily on the system's ability to collect, process, and distribute information rapidly and accurately [1]. The findings regarding the importance of multi-actor coordination also reinforce the common operating picture concept, where cross-institutional data integration is key to improving decision-making efficiency [25]. Thus, the findings confirm that an integrated technology-based approach is a fundamental requirement in modern emergency response systems.

Compared to prior studies, the findings demonstrate consistency with research emphasizing the importance of mobile technology utilization in improving emergency response. Previous studies show that mobile-based applications can improve response speed and accuracy compared to conventional methods [27]. Other research also indicates that geolocation and real-time tracking systems can enhance coordination and operational efficiency in

disaster management [28]. However, unlike previous research tending to focus on a single technological aspect, the findings show that multi-feature integration such as biometric data, incident classification, and volunteer involvement becomes a key factor in improving overall system effectiveness. This indicates a need to develop more holistic approaches compared to partial systems [29].

From a theoretical perspective, these findings contribute to the development of the ICT-enabled emergency response systems concept, particularly regarding multi-actor integration and real-time data. The study expands the understanding that system effectiveness is determined not only by the technology employed, but also by the system's capability to simultaneously integrate diverse data sources and actors. This aligns with the development of AI-driven disaster management concepts, which emphasize the importance of data integration, predictive analytics, and automated coordination in improving response quality [30]. The findings regarding volunteer involvement also enrich the digitalized co-production concept, where collaboration between communities and agencies becomes a critical element in technology-based emergency response systems [5]. Thus, this study expands existing conceptual frameworks by emphasizing integration, collaboration, and real-time data as the primary pillars of emergency response systems.

The practical implications indicate that developing a mobile-based emergency response system like SiagaKita can provide significant benefits for various stakeholders, including government, disaster management agencies, and the general public. Implementing a system capable of providing GPS-based automatic reporting, real-time tracking, and user data integration can enhance resource mobilization efficiency and significantly reduce response time [16]. Additionally, AI and IoT-based communication technology can improve coordination quality and decision-making in emergency situations [31]. From a policy perspective, the findings highlight the importance of integrating emergency response systems at the national scale, as well as the need for regulations supporting interoperability between agencies and user data protection. This also aligns with findings that resilient communication infrastructure is a key factor in successful disaster management [7].

Despite its significant contributions, several limitations must be noted. First, this study uses a qualitative approach with a limited number of informants, so generalization of findings remains contextual. Second, testing of the developed system remains at the conceptual stage and has not been conducted at scale, so system effectiveness in real conditions requires further testing. Third, infrastructure limitations such as network access in remote areas also pose challenges for real-time system implementation. Additionally, user data privacy and security are important aspects requiring attention in personal data-based system development. Therefore, future research is recommended to adopt a mixed-methods approach with empirical prototype testing, integrating data security and infrastructure sustainability as components of a more comprehensive system design [32].

5. Conclusion

This study aimed to design and develop an integrated mobile-based emergency response system, SiagaKita, using a qualitative Research and Development (R&D) approach. This approach enabled an in-depth exploration of the needs of users, volunteers, and agencies within the emergency response ecosystem, producing a contextual and applicable system design. By leveraging primary data from interviews and observations, supported by scientific literature review, this study focused on strengthening coordination, information accuracy, and response efficiency in emergency situations, particularly in the tourism sector with high vulnerability to disaster risks.

The findings indicate that the primary problems in current emergency response systems include fragmented communication, delayed responses due to lack of data integration, and users' limitations in accurately conveying information during emergencies. This study addresses the research questions by demonstrating that mobile-based systems integrating GPS-based automatic reporting, incident classification, real-time tracking, and multi-actor coordination can significantly enhance emergency response effectiveness. Furthermore, user data integration including health condition information—and volunteer involvement within the system have proven to be critical elements in accelerating initial treatment and improving overall emergency service quality.

The scientific contribution of this study lies in the development of a conceptual model and technology-based emergency response system design that integrates various actors and data sources within a unified ecosystem. The study enriches the information systems and disaster management literature by affirming the importance of a holistic approach combining technology, user behavior, and agency operational needs. The practical relevance of these findings is significant for policy makers and system developers in designing adaptive, responsive, and field-needs-

driven emergency response platforms. Thus, this study contributes not only to theoretical development, but also provides real implications for improving emergency response system quality in Indonesia.

Nevertheless, this study has limitations, particularly in the scope of informants and the absence of large-scale field implementation testing. Therefore, future research is recommended to develop and test system prototypes at a larger scale using a mixed-methods approach, to quantitatively measure system effectiveness while maintaining qualitative analytical depth. Future research should also explore data security aspects, inter-agency system interoperability, and technology adaptation in areas with infrastructure limitations, ensuring that digital-based emergency response system development can be more inclusive, sustainable, and widely implementable.

Appendix A. System Module Structure

Table A1. Application Module Structure

Module	Function	Target User
Auth	Login, registration, verification	All users
Community	SOS, reporting, guide, map	General public
Volunteer	Operations dashboard, missions	Volunteers
Agency	Monitoring & coordination	Agencies (future)
Admin	System management	Admin (future)

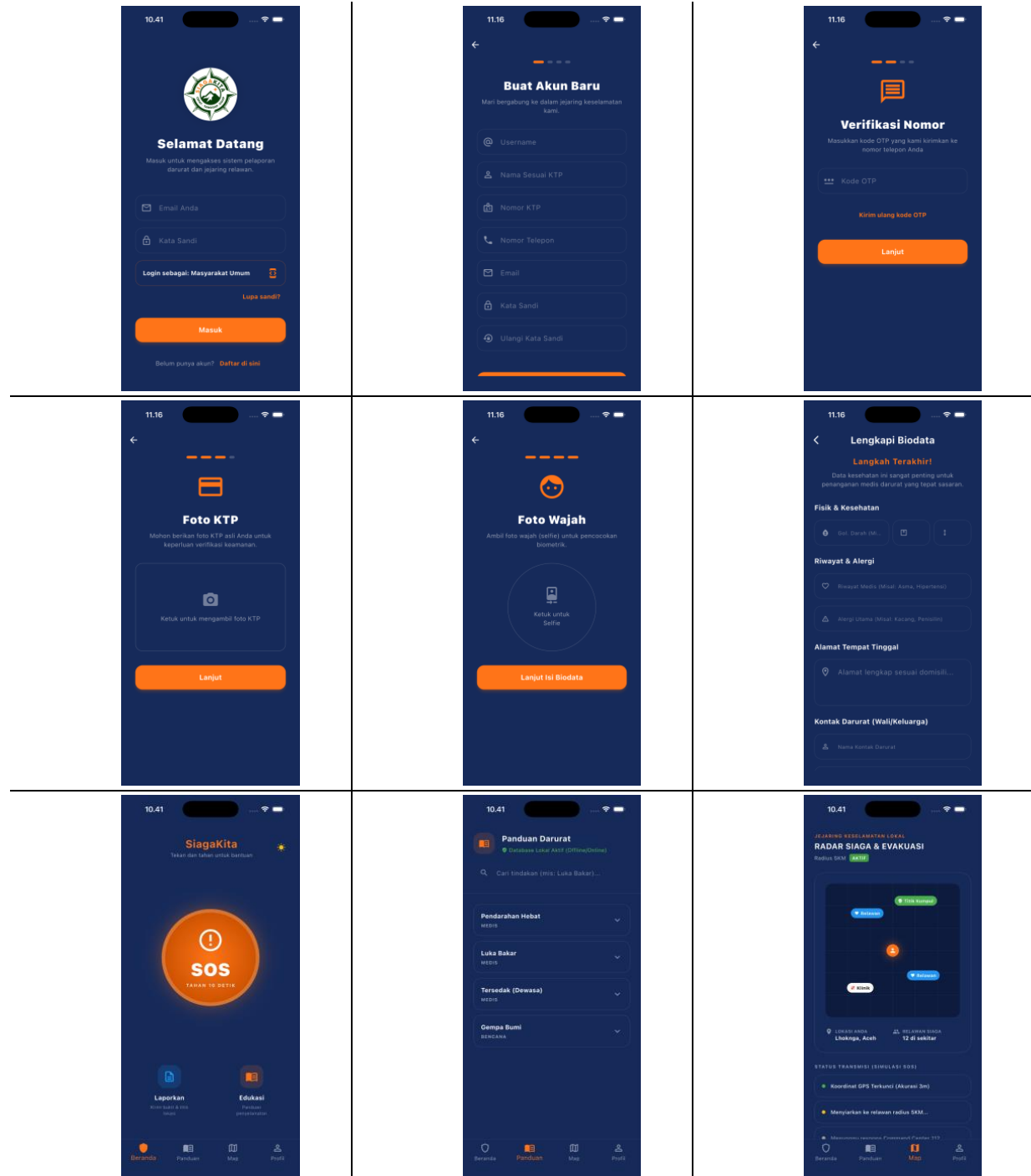
Table A2. Features and API Integration

Feature	API Endpoint	Output
Login	/api/auth/login	Token + user role
SOS	/api/sos/trigger	SOS ID + status
Report	/api/report/create	Report ID
Map	/api/map/nearby	Volunteer & facility data
Profile	/api/user/profile	Complete user data
Volunteer	/api/volunteer/dashboard	Status & missions

Table A3. User Roles

Role	System Access	Characteristics
Community	SOS, report, guide	General users
Volunteer	+ Operations & missions	Field responders
Agency	Monitoring	Decision makers
Admin	System control	Platform management

Appendix B. SiagaKita Application User Interface



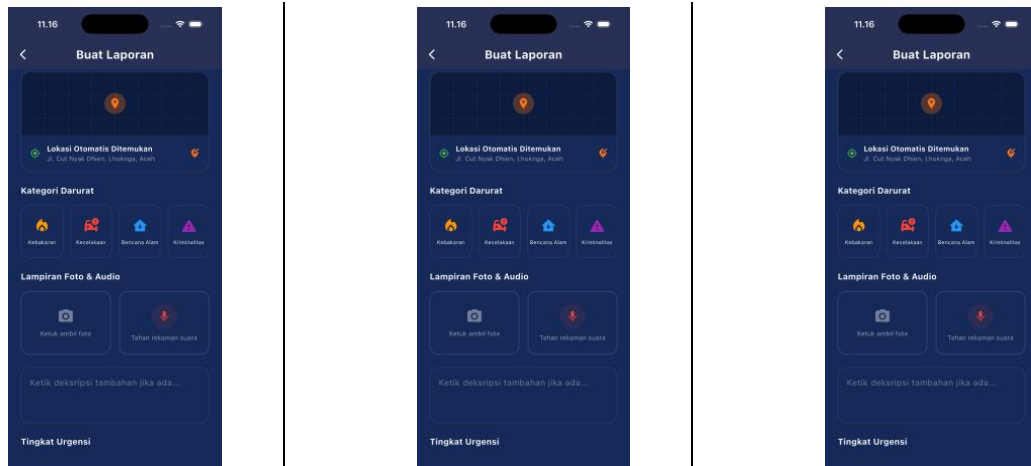


Figure B1. Main UI screens of SiagaKita (login, SOS dashboard, reporting, education, map)

Appendix B presents the SiagaKita application interface designed using the Flutter framework with a user-centered design approach. The login page is kept deliberately simple to accelerate user access in emergency conditions. Upon authentication, users are directed to the main dashboard featuring a central SOS button activated by a 10-second press-and-hold mechanism. The dashboard also provides quick access to the "Report" and "Education" features, directing users to incident reporting and safety guidance respectively. The guidance page supports offline access, while the map page displays volunteer distribution, health facilities, and evacuation points based on the user's location.

Appendix C. User Profile and Medical Data Display

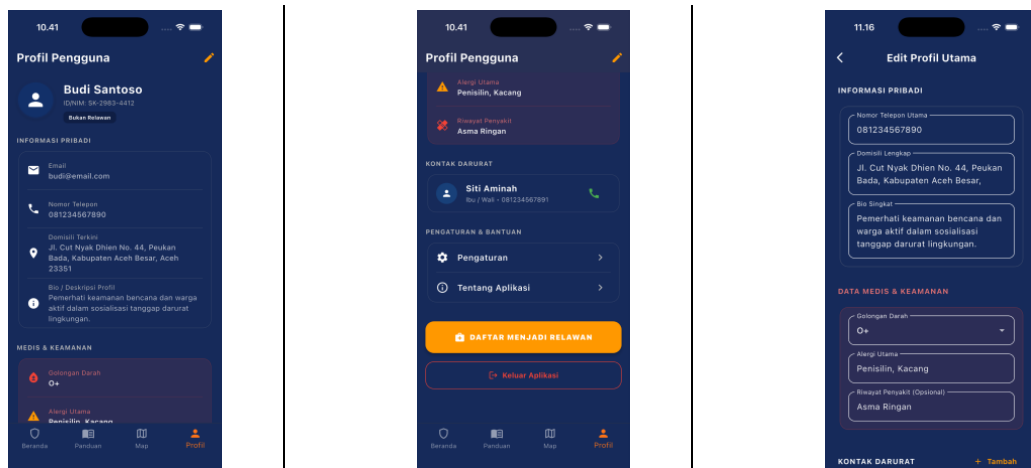


Figure C1. User profile screens with integrated medical data (blood type, allergies, emergency contacts)

Appendix C presents the user profile system integrated with critical medical data, including blood type, allergies, and medical history. This data functions as a biometric emergency ledger accessible by volunteers or medical personnel during emergency conditions. An emergency contact feature further enables rapid communication with the user's family or guardian. This system is designed to improve the speed and accuracy of medical treatment in the field.

Appendix D. Volunteer Dashboard (SiagaKita Tactical Mode)

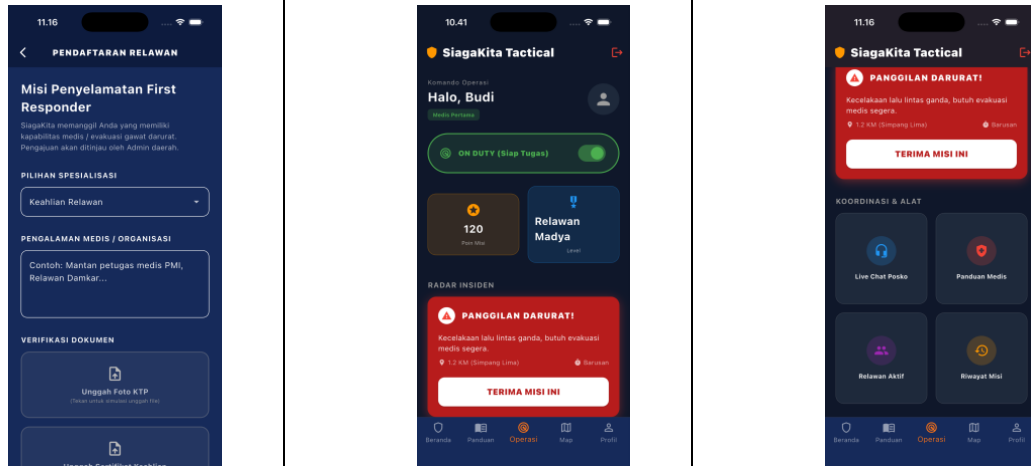


Figure D1. Volunteer-exclusive dashboard with role-based routing, readiness status, and mission system

Appendix D presents the volunteer-exclusive dashboard, displayed only for users with a volunteer role consistent with the role-based routing system where the application interface adapts to the user's role. Key features include readiness status (On Duty / Off Duty), a gamification system with points and levels, an emergency incident radar, real-time mission notifications, and a quick-action "Accept This Mission" button. The navigation bar's "Operations" menu is the primary visual differentiator between general users and volunteers.

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